CADILLAC FORMULA 1 TEAM

2026 Miami Grand Prix — Race Strategy Report
Round 4 | Miami International Autodrome | Sprint Weekend

TEAM SNAPSHOT

	Sergio Perez (#11)	Valtteri Bottas (#77)
Season Best	P17 (Japan — on lead lap)	P13 (Australia)
Current Standing	P19 in WDC	P20 in WDC
Tyre Strength	Elite management, long stints	Technical feedback, consistency
Japan Strategy	Medium start	Hard start (underperformed)
Miami Upgrade	First aero package confirmed for this weekend	

WEEKEND WEATHER FORECAST

Source: Open-Meteo API | Miami Gardens (25.958°N, 80.239°W) | Fetched live

Session	Air Temp	Track Temp	Humidity	Rain %	Wind
Practice 1	31.6°C	52.6°C	40%	3%	14.8 mph
Sprint Race	32.9°C	53.9°C	36%	4%	10.7 mph
Qualifying	26.7°C	34.7°C	58%	6%	11.7 mph
Grand Prix	26.2°C	34.2°C	75%	29% ■	10.1 mph

TYRE COMPOUND ALLOCATION — MIAMI 2026

Pirelli confirmed allocation: C3 (Hard) / C4 (Medium) / C5 (Soft). Softest available range chosen for Miami's smooth, thermally demanding surface.

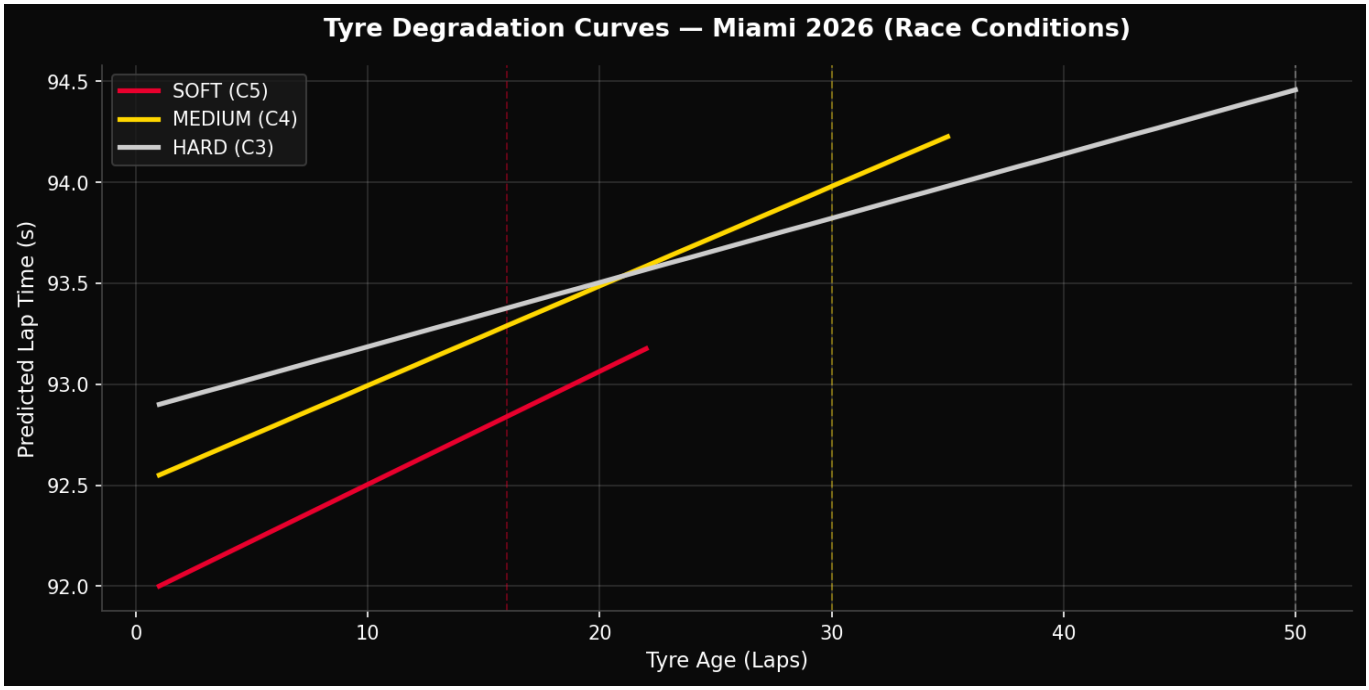
Compound	Pirelli	Pace Delta	Deg Rate (adj.)	Thermal Cliff	Max Stint
SOFT	C5	Baseline	0.0560s/lap	Lap 16	~18 laps
MEDIUM	C4	+0.55s	0.0493s/lap	Lap 30	~35 laps
HARD	C3	+0.90s	0.0318s/lap	Lap 50	~57 laps

Deg rates calibrated from 3,369 clean laps of FastF1 Miami data (2022–2025), then adjusted +7% for 2026 narrower tyre construction and track temperature delta.

TYRE DEGRADATION ANALYSIS

DEGRADATION CURVES — MIAMI 2026 RACE CONDITIONS

Predicted lap times per compound based on FastF1-calibrated deg rates. Dashed vertical lines mark estimated thermal cliff points where deg accelerates. Base pace set to 92.0s for visualisation. Track temp: 36.6°C (race night forecast).



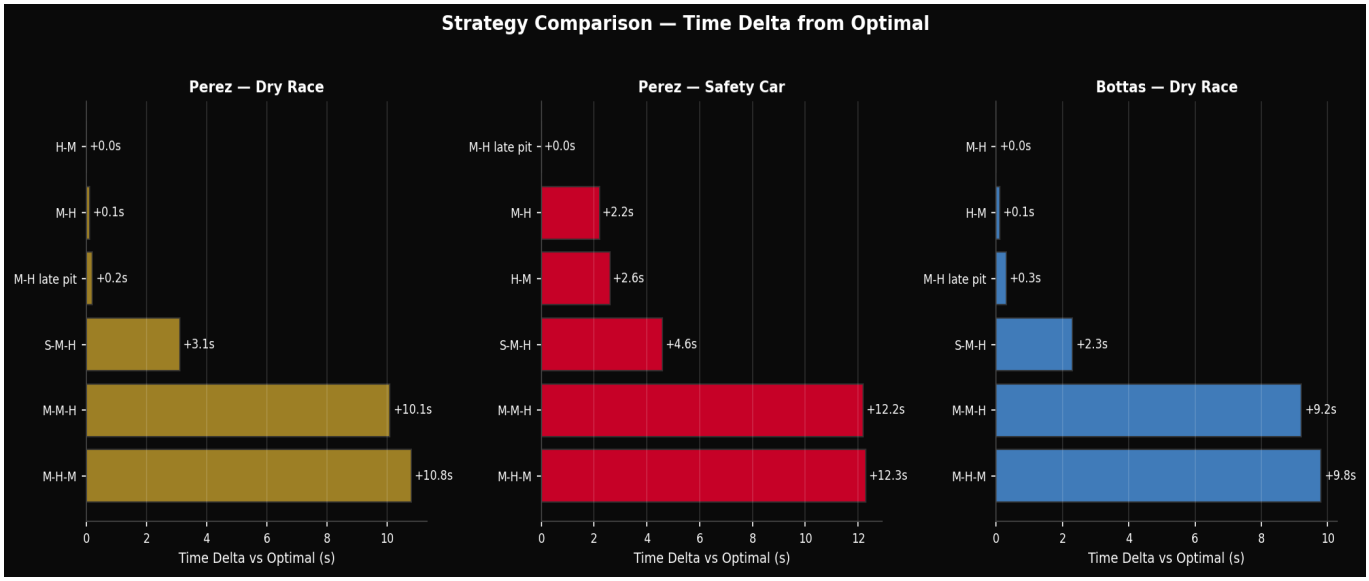
KEY OBSERVATIONS

1. Miami favours 1-stop strategies in all four previous editions. 28 of 30 points-scoring finishes from 2022-2025 were 1-stoppers.
2. The C5 Soft is fastest early but reaches its thermal cliff by lap 16 in Miami's heat. Useful for qualifying simulation and SC restarts only.
3. The C4 Medium is the workhorse compound — preferred opening stint tyre for 15 of 20 drivers in 2025. Manages heat well up to ~30 laps.
4. The C3 Hard shows remarkably flat degradation under race conditions and is the preferred closing stint tyre for 1-stop strategies.
5. 2026 regulation adjustment: narrower tyres (25mm front, 30mm rear) reduce contact patch area, increasing thermal load per unit area. A 7% upward multiplier is applied to all historical deg rates.
6. Race night conditions (36.6°C track vs 52°C daytime) significantly reduce thermal deg vs Sprint Race. 1-stop strategies become more viable in cooler evening temperatures.

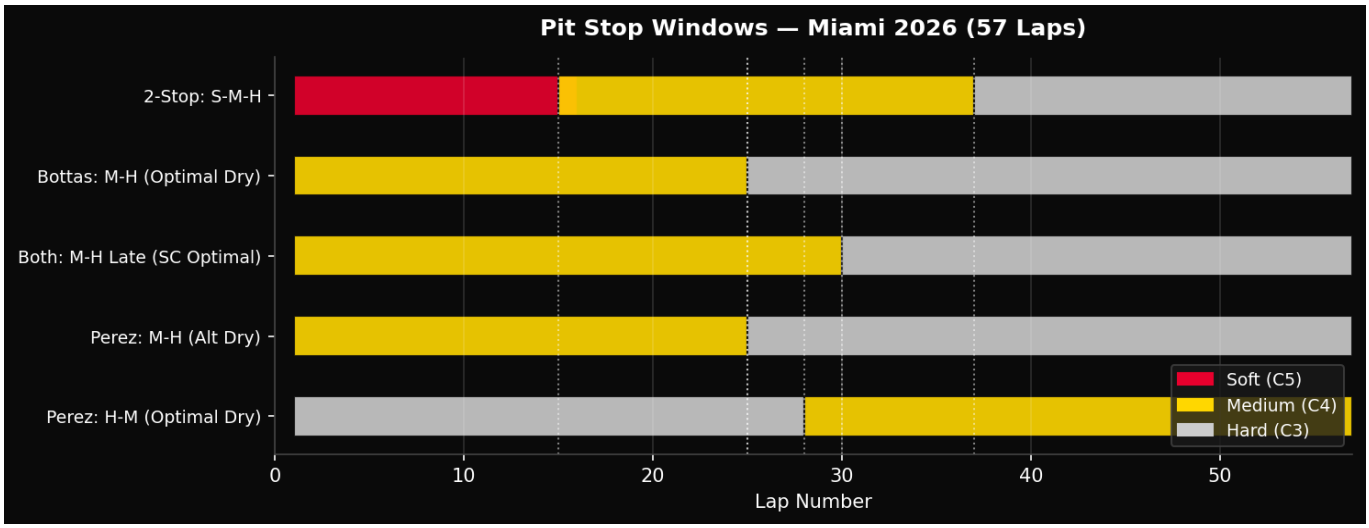
RACE STRATEGY RECOMMENDATIONS

STRATEGY SIMULATION RESULTS

57-lap race simulation run for all viable compound combinations. Pit loss: 22s. Perez deg advantage: 10% reduction applied to all compounds. SC scenario assumes deployment laps 28-31 (consistent with Miami history).



PIT STOP TIMING WINDOWS



RECOMMENDED STRATEGIES

Driver	Scenario	Recommended Strategy	Pit Lap	Rationale
Perez	Dry Race	Hard → Medium	Lap 28	Perez tyre management extends Hard stint. Fresher Mediums in traffic.
Perez	Safety Car	Medium → Hard (late)	Lap 30+	Extend Medium stint, pit under SC to emerge on fresh Hards.
Bottas	Dry Race	Medium → Hard	Lap 25	Classic Miami 1-stop. Hard carries to the flag with flat deg.
Bottas	Safety Car	Medium → Hard (late)	Lap 30+	Mirror Perez — opportunistic SC pit maximises position gain.

Both	Rain (29% risk)	Pit for Inters immediately	On request	100% cloud cover race night. First team to react gains 5-8 positions.

RISK & OPPORTUNITY MATRIX

Factor	Probability	Impact	Cadillac Action
Safety Car	High (65%)	High	Stay out if on Medium, pit immediately if on Hard
Rain during race	Medium (29%)	High	Pre-position Inters on pitlane wall from lap 20
Rival 2-stop bluff	Medium	Medium	Hold 1-stop unless deg cliff materialises early
Upgrade underperforms	Low	Medium	Revert to conservative pace management
Track evolution	Certain	Low	Expect 0.5-1.0s natural improvement by lap 20

Report generated programmatically using FastF1 telemetry data (2022-2025), Open-Meteo live forecast API, and a linear tyre degradation simulation model. Assumptions: 22s pit loss, 57-lap race distance, Perez 10% deg management advantage. Author: Arnav Timmapur | github.com/Arnavgt-04/2026-Miami-Race-Strategy-Cadillac